

On structural parallelism in formal music semantics

Léo Zaradzki

Lycée Saint-Louis, Paris, France
leozaradzki@gmail.com
ORCID iD: 0000-0001-5920-2318

Abstract

Philippe Schlenker’s recent semantic system for music states formal necessary conditions for a musical snippet $\mathcal{M}$ to denote a given situation $\mathcal{S}$. According to these conditions, some features of the music and the scene must evolve in a parallel way through time. In this article, I raise the question of the syntax–semantic interface in music, which has not been investigated in previous works. I argue that Schlenker’s “linear” conditions are not sufficient and that the denotation relation also obeys some structural conditions: both $\mathcal{M}$ and $\mathcal{S}$ exhibit a tree-structure and these two structures must match in a way or another. After investigating original examples showing that structural conditions are needed, I present an assortment of such conditions in the formalism of rooted trees. Some of these conditions constrain the trees $\mathcal{M}$ and $\mathcal{S}$ in a symmetric way (meaning that $\mathcal{M}$ and $\mathcal{S}$ play the same formal role), while others are asymmetric, and both possibilities are investigated. Finally, I examine logical and entailment links between the different conditions stated, leaving future corpus and empirical studies to decide which is the most relevant.

Keywords : supersemantics, music semantics, syntax–semantics interface, musicology

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1 Introduction

The idea that music conveys a meaning is as controversial as it is ancient. Through the twentieth century alone, many philosophers and musicologists discussed it. There is a consensus on the fact that music *does* convey meaning, and there is another consensus on the fact that “meaning” in this assertion is not to be understood in the same way as when we say that language conveys meaning. The controversy is not focused on these questions but rather on the nature of this musical meaning. The main historical positions are expounded in the first chapter of *Emotions and Meaning in Music* (Meyer, 1956). The most important confrontation is between absolutists, who think that the meaning of music is only intramusical, and referentialists, who think that music can refer to the outer world. Inside these two camps, several positions coexist¹.

After the development of formal semantics for linguistics during the second part of the twentieth century, several formal systems have been proposed for musical semantics as well (Steedman, 1996; Granroth-Wilding et Steedman, 2014; Larson, 2012; Schlenker, 2017, 2019). The particularity of Schlenker’s formal system is that it is entirely and explicitly extramusical: “by ‘semantics of music’ we mean a rule-governed way in which music can provide information

¹I refer to Meyer (1956) and to others (Pratt, 1938; Koelsch, 2012) for the philosophical discussions on this point.

(*i.e.* license inferences) about some music-external reality, no matter how abstract” (Schlenker, 2019). Schlenker proposes a few formal rules as necessary conditions for a musical snippet $\mathcal{M}$ to denote an extramusical situation $\mathcal{S}$ (thereafter written $\mathcal{M} \models \mathcal{S}$). However, Schlenker’s system is still young and, as the title of his article indicates, it was not designed to propose an exhaustive semantic system but to provide a paradigm and a framework in which formal rules could eventually be formulated.

In this article I elaborate on one aspect of Schlenker’s semantics, which can be called structural parallelism. As has been argued many times, both music and situations in the real world can be assigned mereological structures, *i.e.* organised into parts and subparts (see more details below). These structures directly or indirectly reflect cognitive parsing by listeners/viewers and can be derived from Gestalt principles. Schlenker put forward the idea that a correspondance between the structures assigned to $\mathcal{S}$ and $\mathcal{M}$ was a necessary condition in order to have $\mathcal{M} \models \mathcal{S}$. Nevertheless, this general principle of what could be called “parallelism” was not made precise in his foundational article (Schlenker, 2019, §8). In a case-study on walk-denoting music, Migotti and I have proposed some elementary formalisation of this idea in terms of rooted trees, in order to discuss more precisely theoretical questions raised by that study (Migotti et Zaradzki, 2019). In this article the authors posited that the structural condition weighing on $\mathcal{M} \models \mathcal{S}$ was that the function f associating situational events of $\mathcal{S}$ with their musical representations in $\mathcal{M}$ should be an embedding from $\mathcal{S}$ to $\mathcal{M}$, that is, should preserve the grouping structures. As I will show below, although this first condition was enough to express and discuss the basic questions the authors wanted to raise in that article (including the question of the status of semantically vacuous notes in $\mathcal{M}$), it does not exactly capture the intuitive notion of parallelism that can be inferred from musical data, and should be refined.

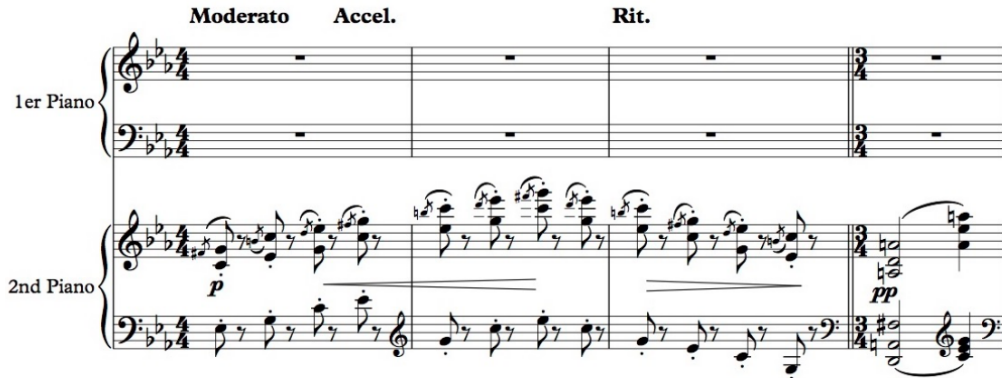
In this article I propose to refine these notions by: a) formulating the relevant mathematical framework of rooted trees in which structural conditions can be easily formulated, b) stating several conditions of parallelism, stronger or weaker, that may govern the denotation relation and, c) based on the framework of rooted trees, examining the logical links between these different conditions. By providing and examining several structural conditions rather than just one, the goal of this article is to allow future research to discuss with much more precision, based on corpus studies and experimental data, the different shades of parallelism that may rule musical semantics in a given context. Throughout the article, I shall illustrate the different concepts by relevant musical examples.

2 Schlenker’s formal semantics of music

In this section, I will describe the general picture of Schlenker’s system, focusing on the general framework (on which the subsequent development elaborates), rather than the so-called truth-conditions which are the system’s actual innovative centre, but whose details are independant from the point of the present article. To see these details, the reader is referred to Schlenker’s pioneer articles (Schlenker, 2017, 2019), from which the following elements are extracted.

2.1 Source-based semantics

The first point that needs to be clearly understood is that Schlenker’s semantics is a *source-based* semantics. This feature distinguishes it radically from usual semantic theories used for natural languages, for those semantics are compositional. This means that in natural language semantics, one considers that there first exists a lexical semantics, which assigns to every meaningful unit — to make it simple: to every word — a meaning (*e.g.* a reference). Then syntax

Figure 1: Saint-Saëns’ *Kangourous* — score reproduced from Schlenker (2019)

makes it possible to form sentences from these meaningful units. The meaning of a sentence is understood as a function of its different parts’ meaning, and the role of compositional semantics is to explain the mechanisms by which those functions act.

A source-based semantics as Schlenker’s is fundamentally different from those compositional systems, for it is not stated compositionally. One can see a musical piece as a sequence of musical events² $m_1, \dots, m_n$, which will represent the *traces* or *projections* of a situation involving one or several *virtual sources*. The word “projection” here is to be understood as in “orthogonal projection”, onto the “acoustic space”. The virtual source (or denotation) is called so because it is not the actual source of the sound, but rather some abstract agent in an abstract space, and the sound is a depiction of its actions. To take a simple example, the first phrase of Saint-Saëns’ *Kangourous* (figure 1) *may* describe a situation involving a jumping kangaroo. Every beat (that is, every musical event) represents another jump of the kangaroo³. Here the kangaroo is the virtual source described by music, but it is of course not the source of the sound. Thus a piece of music depicts the virtual source in some situation, and the musical events reflect the events that constitute this situation. The meaning of a musical piece is not derived from the meaning of its subunits as in language. Rather the music describes “in real time” the situation unfolding under our ears.

The second very important point is that the situation denoted by a musical piece is of course not unique. The music on figure 1 might describe plenty of things besides kangaroos. This is actually not different from what happens with language, since the sentence “a kangaroo is jumping” likewise does not refer to a specific kangaroo in a specific environment, but to a generic kangaroo in a generic environment, and different people would imagine slightly different mental scenes after hearing that sentence. Just the same way, after hearing the music from figure 1 some people may imagine a kangaroo jumping while others might imagine a dog barking, but the diversity of these situations does not mean that the music has no meaning, but only that musical meaning is broader and looser than linguistic meaning. On the contrary, there are numerous situations that this same piece could *not* depict properly, and this indicates that the notion of musical meaning makes sense. A precise definition of the meaning of a piece has been proposed by Schlenker: just as in philosophy of language, the meaning of $\mathcal{M}$ is the set of all its

²I shall not discuss the notion of musical events here, and I refer to my Phd dissertation for a discussion of it (Zaradzki, 2021, chap. 4); in first approximation, we shall see every note or chord in a piece as a separate musical event

³Of course this situation is strongly suggested by the piece’s title. Hence we would say that the music *may* depict this situation, and not that it *does* necessarily depict it. More on this central point below.

The image shows a musical score for Saint-Saëns' *Éléphant*. It consists of two staves. The top staff is for the 2d Piano, marked *Allegretto pomposo* and *f*. It features a series of chords in the right hand and single notes in the left hand. The bottom staff is for the Contrebasse, also marked *Allegretto pomposo* and *f*. It features a series of single notes in the right hand and chords in the left hand. The key signature is three flats (B-flat, E-flat, A-flat) and the time signature is 3/8.

Figure 2: Saint-Saëns' *Éléphant* — score reproduced from Schlenker (2019)

possible denotations $\mathcal{S}$:

$$\llbracket \mathcal{M} \rrbracket = \{ \mathcal{S} \text{ situation}, \mathcal{M} \models \mathcal{S} \}$$

and now the burden has shifted on the denotation relation $\models$, which I describe further below⁴.

2.2 Musical parameters, semantic inferences and musical meaning

Schlenker's main idea was to note that music is able to convey information about the world because certain musical parameters (including timbre, height, loudness or harmonic stability) have a semantic interpretation, and provide the listener information either on what the source does or on what it is.

To cite another fine example from Schlenker (2019), let us consider Saint-Saëns' *Éléphant* on figure 2. The double bass' deep timbre seems to trigger the inference that the virtual source has a big size (for instance, an elephant). To test this, Schlenker used the method of minimal pairs. Playing the melody two octaves higher removes the inference of a big animal, and playing it one more octave higher triggers an inference of a small animal⁵.

In fact, this inference can be explained by general auditory cognition principles: bigger things — and especially bigger animals — tend to produce deeper sounds⁶.

⁴As Regine Eckardt perceptively pointed out to me, the notation $\mathcal{M} \models \mathcal{S}$ is opposite to model theory's, where one would write $\mathcal{M} \models \varphi$ to state that some structure $\mathcal{M}$ is a model of the formula φ : the signified is on the left while the signifier is on the right. In Schlenker's notations, the signifier (the music) is on the left and the signified on the right. While this has no consequence on the theory as it is just a matter of notation, the reader may prefer to replace $\mathcal{M} \models \mathcal{S}$ by $\mathcal{S} \models \mathcal{M}$ from beginning to end. I will personally use Schlenker's original notation to avoid confusion.

⁵At this stage of research this inference still rests on introspection, but some of them have been tested experimentally and in these experiments, musical introspection has been corroborated (see for example Migotti et Zaradzki (2019)).

⁶As Schlenker notices, this inference is so strong that certain animals apparently evolved mechanisms to lower their vocal-tract resonant frequencies so as to exaggerate their perceived size (Fitch et Reby, 2001).

The major part of Schlenker’s article is devoted to highlighting semantic effects by the minimal pairs method. They can be classified in two categories: those resulting from general auditory cognition, as the effect triggered by the pitch in the *Éléphant*, and those that are specific to musical cognition, that is related to tonal or harmonic properties of music: for example the less harmonically stable the chords are, the less physically stable the source is perceived by the listener.

I shall not go through all the semantic effects highlighted by Schlenker here, since the list would certainly not be exhaustive. I refer rather to Schlenker’s aforementioned articles for a list and for evidence. Let us just emphasise that the core idea of Schlenker’s semantics is that some musical parameters are associated with specific kinds of inferences on the virtual source, and that the evolution of these musical parameters throughout the musical piece will depict the source’s adventure (which can be very abstract indeed). For instance, one of the semantic effects demonstrated by Schlenker is that increasing loudness may mean that the virtual source is coming nearer.

One can thus conclude that in order to have $\mathcal{M} \models \mathcal{S}$, the chronological evolution of each musical parameter throughout $\mathcal{M}$ should reflect the chronological evolution of its denoted property throughout $\mathcal{S}$. To put it differently, $\mathcal{S}$ should reflect the linear evolution of $\mathcal{M}$ regarding all the musical parameters (see [Schlenker \(2019\)](#) for a more complete but still maybe non exhaustive list). Formally, if $\mathcal{M}$ is to be decomposed into musical events $m_1, \dots, m_n$, and $\mathcal{S}$ into events $e_1, \dots, e_n$ (more on this below), and if we note ℓ the loudness function, d the distance function⁷, and g the function representing the energy of the source⁸, then:

Condition 2.1 (Schlenker, 2019). For $\mathcal{M} \models \mathcal{S}$, it is necessary that:

$$\forall i, j \in \{1, \dots, n\}, \ell(m_i) \leq \ell(m_j) \implies d(e_i) \geq d(e_j) \text{ or } g(e_i) \leq g(e_j)$$

To have a complete picture of the $\models$ relation, one has to add analogous conditions for all the musical parameters. Thus, a possible denotation of a musical piece $\mathcal{M}$ is any situation $\mathcal{S}$ that would fit to $\mathcal{M}$ regarding all the different musical parameters; that is, a possible denotation of $\mathcal{M}$ is any situation that is *isomorphic* to $\mathcal{M}$, the word “isomorphic” concerning here the order relations on all the musical parameters and their semantic counterparts. Finally the meaning of $\mathcal{M}$ is the set of all these isomorphic situations.

As a reminder, one of the goals of this article is to show that these linear conditions on musical parameters are necessary but not sufficient for $\models$, and other conditions regarding the structure play a role as well. This is what I will argue in section 4. Before explaining what is meant by “structure” in section 3, I shall conclude the present section with a short digression on the nature of musical meaning.

2.3 Digression on the nature of musical meaning

The definition:

$$[[\mathcal{M}]] = \{\mathcal{S} \text{ situation}, \mathcal{M} \models \mathcal{S}\}$$

is analogous to that of linguistic meaning. Still, it has been noticed that the meaning of a musical piece is far wider or more abstract than the meaning of a sentence in natural language. Does it mean that the $\models$ relation in music is of a different nature? To conclude this section on Schlenker’s semantics, I shall list some elements shedding light on this question.

⁷Distance from the source to the auditor in the virtual space in which the situation takes place.

⁸Indeed, Schlenker connects loudness to both distance and energy; see details in the aforementioned article.

When we say it is raining, our interlocutor is immediately able to build a precise and un-equivocal mental representation of this statement. As for music, the process seems slightly different: taking a musical stimulus as an input, our brain does not seem to be able to spontaneously build such a mental representation. Nevertheless, as soon as the brain receives as an input both the musical stimulus *and* a potential denotation, the task of determining whether or not the stimulus and denotation correspond is easier and more natural, especially when the denotation is compared to another, less relevant, one. In other words, out of the following two tasks, we seem able to deal with the second but not with the first:

1. Given $\mathcal{M}$, find a situation $\mathcal{S}$ such that $\mathcal{M} \models \mathcal{S}$;
2. Given $\mathcal{M}$ and $\mathcal{S}$, decide if $\mathcal{M} \models \mathcal{S}$.

This way of putting things shall remind logicians of the problems of decidability and proof verification. For a computer, the problem of deciding if a statement is demonstrable in a system of axioms is much more difficult — if not impossible — than the problem of deciding if a given proof of this statement is valid. Of course this is because exploring all the possible proofs (that is, sequences of formulae leading to the final statement) takes more than exploring the deduction rules to check each step of the given proof, since the number of such rules is very limited.

It seems to me that the phenomenon is similar when it comes to music and its denotations, for the number of situations potentially matching with a piece of music $\mathcal{M}$ is too large for the brain to pick one among the others. This would be corroborated by the idea that when the listener is guided — by the title for instance — to imagining a given virtual source, they build mental representations more easily.

The size of the set of possible denotations is the reason why Schlenker says that musical semantics is abstract: since they are so numerous, the nature of the greatest common divisor of all the denoted situations is too abstract — for instance “the virtual source gains energy or comes closer, and gets from an instable state to a stable state” — for the listener to be able to visualise particular examples of it. Abstract or not, that still remains a *meaning* for music, as some remarks in Meyer (1956)’s Chapter 1 echo.

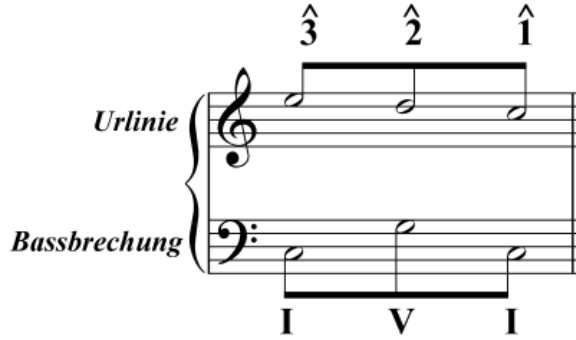
3 Structures on music and situations

In a previous case study on walk-denoting music, Migotti and I developed the hypothesis, previously evoked by Schlenker (2019, §8), that the denotation relation $\models$ may be ruled not only by linear conditions as the isomorphism conditions seen in the previous section, but by structural conditions too (Migotti et Zaradzki, 2019). What is meant by structural conditions is that both a situation $\mathcal{S}$ and a musical piece $\mathcal{M}$ possess a structural organisation of their parts, and that both structures matching is a necessary condition in order for $\mathcal{S}$ to denote $\mathcal{M}$.

In this section, I shall recall evidence from previous work that these structures exist and are naturally perceived by listeners/viewers, and then I will explain in some detail what they are and how they can be represented.

3.1 Musical structures

On the musical side, the idea of hierarchical structures on musical pieces dates back at least to Heinrich Schenker and the so-called schenkerian analysis, developed at the beginning of the

Figure 3: Schenker's *Ursatz* in C

twentieth century. Many authors have then elaborated on and formalised his ideas, so I will just make it short and refer to their work for further details.

According to Schenker, all the musical pieces of occidental tonal music are derived from a very limited set of fundamental patterns. Each of these fundamental patterns constitutes a skeleton, on which the final piece is built by recursively applying so-called *elaboration* rules. For instance, arpeggiating a chord or adding passing notes are commonly used elaboration processes. So Schenker's central idea is that every tonal piece is in a theme and variations relationship with one of those fundamental patterns. The most common fundamental pattern, called the *Ursatz*, is shown in figure 3. This is the deep structure associated with most tonal pieces, which are the surface thereof. Since this article is not meant as an introduction to schenkerian analysis, we refer to Pankhurst (2008), a textbook addressed to musicians.

Now the fact that every musical piece is, according to this theory, a recursive elaboration on this *Ursatz* means that every musical piece has a hierarchical structure, as has every recursive object. Several formalisations thereof have been proposed, the most famous of which remains — though there is more than that in it — Lerdahl and Jackendoff's pioneer *Generative Theory of Tonal Music* (thereafter GTTM, Lerdahl et Jackendoff (1983))⁹. Since our goal is not to determine which exact rules hold in the establishment of such structures, I shall not discuss further the different theories that have been proposed in time. The mere fact that there exists a way in which listeners parse musical pieces into hierarchical structures is sufficient for our purpose.

3.2 Situation structures

Now that we have argued that on the musical side $\mathcal{M}$ is perceived by listeners as mereologically organised, the same question holds on the situation side $\mathcal{S}$. But to begin with, let me clarify what I mean by the term “situation”. Without attempting to give an exact definition, situations here should be thought of as *scenes* happening, and involving an agent¹⁰. It is close to the homonymous concept from situation semantics. One important feature of situations is that, unlike logical utterances for example, they must be chronologically ordered; that is, they unfold over time. That is why it is possible to evaluate them alongside music and determine if both are isomorphic, as described above. I assume situations to be best thought of in the visual modality. Here, the concept of mental imagery seems relevant. According to the definition

⁹GTTM actually offers a range of four different, all cognitively motivated, ways of structuring a musical piece in a hierarchical fashion.

¹⁰Recall that this agent, that Schlenker refers to as “the source”, may be very abstract.

given by the Stanford Encyclopedia of Philosophy, “mental imagery is perceptual representation not triggered directly by sensory input” (Nanay, 2021) — here it would be indirectly triggered by music. Mental imagery in general is not necessarily visual and can correspond to other modalities, but most of the examples cited are visual, and the very name “mental *imagery*” evokes that visual mental imagery is an important part of it. Mental imagery can be triggered by perception. One example cited in the article is when you hear the voice of someone you know with your eyes closed, you see their face. This is an example of multimodal mental imagery, that suggests that the same phenomenon might be at stake in music semantics. According to Nanay (2021) again, “one important set of findings come from the ‘dual coding theory’, according to which linguistic representations themselves are partly constituted (or at least necessarily accompanied) by mental imagery and this explains why concrete words (that are accompanied by more determinate mental imagery) are easier to recall than abstract words”. Moreover, even if mental imagery can involve several modalities, as far as we are concerned here in music semantics, the auditory slot being already occupied by the musical signal, another auditory denotation than the music itself would be hard to imagine, and therefore a visual version of mental imagery seems the most probable. Due to these considerations and also for practical reasons, all the situations described here will be thought of as visual scenes, and even sometimes as videos. I will consider indeed that when some video is synchronised with a musical snippet $\mathcal{M}$, the situation $\mathcal{S}$ shown by the video is one of the many situations denoted by the video, and thus, that both must satisfy all the criterions of the $\models$ relation¹¹.

In a famous series of experiments, Zacks et al. (2001) have shown to participants some daily life video recordings. These situations were either very common activities (such as make the bed, do the washing-up, *etc.*) or more specialised ones (such as assemble a saxophone, reupholster a chair, *etc.*). While they watched the video, participants had been asked to press a button every time they noticed the end of a unit and the beginning of a new one. But whereas one group was instructed to press the button most often (every time a step change was perceived), the other one had to press it less often (only at the coarse boundaries of the activity). Zacks and his colleagues then compared the times at which participants placed boundaries. They did find that the first group’s delimitations were on average finer than the second’s. But above all, they realised that every time that the second group signaled a unit boundary, the first group did too. Therefore subjects perceive situations as sequences of steps organised into mereological hierarchies, *i.e.* situations are sequenced, in the participants’ minds, into steps, and those steps themselves are sequenced into substeps, and so on. The authors also mention that these steps are always *goal-directed*. Variations on this experiment then helped to show that participants sequenced situations the same way when it came to remember or to describe them.

These results suggest that situations, just as music, are perceived as organised in hierarchical structures. Jackendoff (2009), in an article investigating the similarities and differences between language and music, illustrated it using the example of preparing coffee. Figure 4 is reproduced from this article. However, Jackendoff used this example only to show that hierarchical structures were no prerogative of language and music (since they were shared also with situations), but as far as I know no formal system has been proposed to derive situation parsings from axioms. Once again, our goal still is not to determine which exact rules hold in the establishment of such structures. For this reason I shall not propose a formal system for

¹¹Of course, synchronising music with video is an artistic practice, and as such can benefit from breaking the rules for aesthetic reasons: when discussing matches and mismatches in dance and music, Charnavel (2022, p.38) notes that “complete match is very rare: in actual choreographies, total match would be artistically uninteresting; and [...] the modality difference between dance and music makes total match unlikely even in natural occurrences of dance-music”. This presents an additional challenge for us; more on this later.

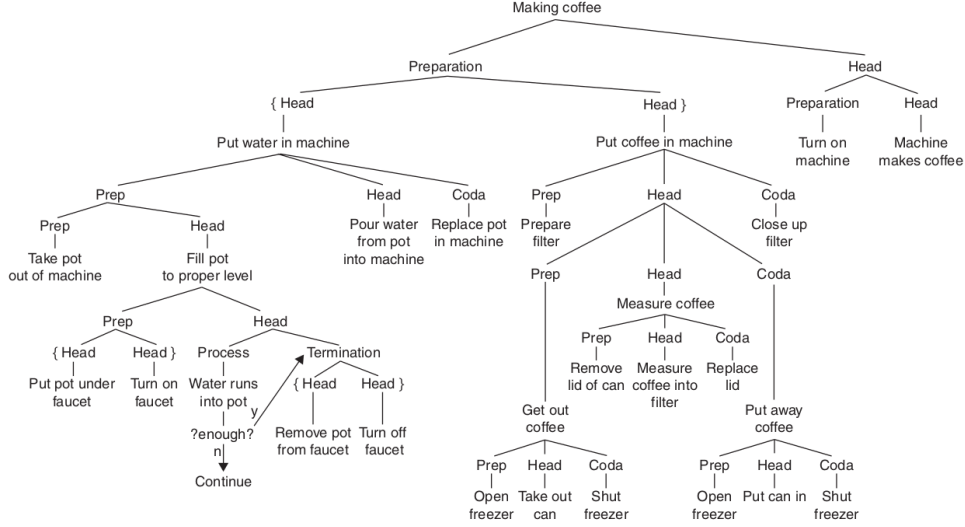


Figure 4: Preparing coffee, according to Jackendoff (2009)

situations here. The mere fact that there exists a way in which viewers parse situations into hierarchical structures, no matter how, is sufficient for our purpose.

Now that I have argued that music and situations are both equipped with perception-based hierarchical structures, the rest of the article deals with the question of the matching between these two structures.

4 Examining structural conditions

Now that I have summed up how hierarchical structures could be assigned both to music and situations, I will focus on the relevance of these structures to the $\models$ relation. As explained in the introduction above, the idea of structural parallelism, that is, the fact that $\mathcal{M}$ and $\mathcal{S}$'s structures should somehow match in order for $\mathcal{M}$ to denote $\mathcal{S}$, is already suggested in Schlenker (2019, §8). Though several leads are discussed there, the Prolegomena are, as their name indicates, still speculative. In particular, little evidence is brought to support the existence of this syntax-semantic interface, and the exact links between $\mathcal{M}$ and $\mathcal{S}$'s structures are not examined in detail¹². This is precisely what I propose to focus on here. First I shall argue, based on musical examples, for the necessity of structural conditions, and then propose a variety of such conditions, without wanting to decide between them at this stage of research. The rest of the article then formalises these different conditions and examines logical links between them. These conditions could eventually be applied to musical semantics as proposed here, but also to numerous situations where two structures are supposed to have a certain degree of matching, like dance-music performances or, on a more syntactic level, congruence between time-span reductions and prolongational reductions.

¹²Case studies showing that interactions between different structures exist have already been conducted — see, among others, Charnavel (2022) about interactions between dance and music. These studies insist on the relevance of match-mismatch relations between structures (also called *congruence* by Lerdahl et Jackendoff (1983)), but formal conditions capturing the intuition about this concept are not investigated. This is what this article proposes to do.

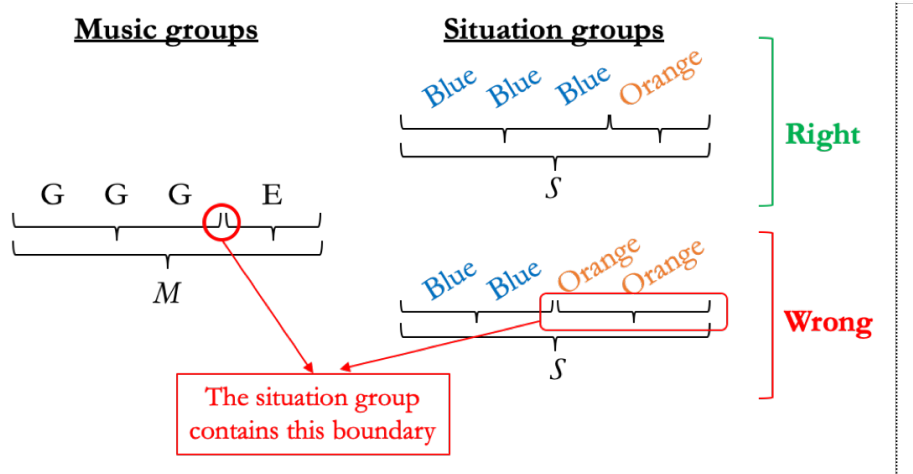


Figure 5: Example of grouping mismatch — reproduced from Migotti et Zaradzki (2019)

4.1 Why structural conditions are needed

Schlenker’s intuition is this: “A musical group A is taken to belong to a musical group B if (on any true interpretation) the world event denoted by A can naturally be taken to be a sub-event of that denoted by B . In other words, grouping structure will be taken to reflect the ‘part-of’ relations among the denoted events, what is called ‘mereology’ (or sometimes ‘partology’) in semantics.” Though I agree with this idea, Schlenker’s article does not provide musical data to support it.

In a subsequent article, Migotti and I have illustrated this principle with a minimal example (Migotti et Zaradzki, 2019, §4.1):

“[Consider] the opening motif of Beethoven’s 5th Symphony. This consists in four notes: G G G E. According to Lerdahl and Jackendoff’s rules they are grouped as $[[G, G, G], E]$. In the corresponding section of *Fantasia 2000*¹³ each note is illustrated by a coloured lightning: the Gs are interpreted as blue lightnings, and the E as an orange one. It seems to us, though, that image and sound fit far less well if we colour, say, the third lightning in orange too. As shown in [Figure 5], this is because the orange group would then contain the G/E boundary.”

In Schlenker’s terms, the mismatch would be caused by the fact that the subgroup $[G\ G\ G]$ of the motif is represented by two blue and one orange lightnings, which cannot be taken to be a subevent of the four lightnings event, which is consistent with Gestalt principles. That one denotation is preferred to another proves at least that structural parallelism plays a role in the $\models$ relation.

Which role exactly, with which constraints and on which priority, is not the subject of this article. Answering these very important questions requires a corpus of minimal pairs of musical snippets together with videos of possible denotations, and experimental data made of participants’ judgements on each of these pairs. This is left for future research. What I can do here is to provide further pieces of evidence based on personal introspection (just as Figure

¹³An animated film where image is intended to be a denotation of the music; see: <https://www.youtube.com/watch?v=nMnlxYkZKaU>

5) and then, in the rest of the article, discuss several shades of structural parallelism and the logical links between them.

4.2 Different shades of structural parallelism — A Charles Chaplin case study

The minimal example from *Fantasia* led Migotti and me to the following speculative condition¹⁴ on the denotation relation:

Condition 4.1. For a musical snippet $\mathcal{M}$ to denote a situation $\mathcal{S}$, the grouping structures of both must match, in the following sense: a group on the situation side must not contain a group boundary on the musical side¹⁵. More precisely, this means that if $e_{i_1}, \dots, e_{i_k}$ ($k \in \mathbb{N}^*, i_1 < \dots < i_k \in \{1, \dots, n\}$) and a are events of $\mathcal{S}$ associated with musical events $m_{i_1}, \dots, m_{i_k}$ and b , and if there is a group to which $e_{i_1}, \dots, e_{i_k}$ belong but a does not belong, then every musical group to which b with some of the m_{i_j} belong, must contain all the m_{i_j} (otherwise this would put up a boundary between the m_{i_j} where there was not between the e_{i_j})

But beginning with the beginning, why not rather posit a strict parallelism between $\mathcal{M}$ and $\mathcal{S}$, such as: groups in $\mathcal{M}$ and groups in $\mathcal{S}$ must strictly match, period? As a control, let us consider a third variation on this example: same musical motif, but with a situation consisting this time in four blue lightnings (therefore organised in one single group). It seems to me that this would be a correct denotation for Beethoven’s motif, overflowing this too strict condition. And indeed this does not violate Condition 4.1 since one cannot find a group in $\mathcal{S}$ to which the first three lightnings belong while the last one does not¹⁶.

Here is another, more elaborated example¹⁷. Let us consider the shaving scene from Chaplin’s *Great Dictator*¹⁸, which is a silent scene inside this speaking film. Charles Chaplin plays a barber who shaves his customer while listening to Brahms’ 5th *Hungarian Dance*. The comical side of this scene lies in the fact that the barber’s actions are precisely in the set of possible denotations of the excerpt, so that he appears to act in synchronisation with the music. The first sixteen bars of the piece serve as an introduction to the scene, as Chaplin prepares his equipment and comes close to the customer. I will focus on bars 17 to 24, whose melody is reproduced on Figure 6 (the analysis of the next eight measures, for example, is more straightforward). This theme is accompanied by beats and off-beats in the orchestra. During this period, Chaplin applies the shaving cream in the following precise way: on first beats he spreads the foam, while on second beats he dunks the brush in the bowl. This is for the first six bars, while for the last two bars, he spreads both on first and second beats, as shown in Figure 6 (where the letter S represents spreading and the letter D dunking). Note that it is no coincidence that the

¹⁴This is very close to rule PRPR2 on prolongational reduction from Lerdahl et Jackendoff (1983), although in that context it is a quite weak preference rule rather than a strict rule. As stated above, though the motivation in this article is different, the formal notion to be developed below could also be applied to the notion of congruence from GTTM, even if it is only syntactic and does not deal with extramusical semantics. The question of whether prolongational reduction might be a form of intramusical semantics is also interesting but will not be investigated here.

¹⁵This wording suggests that situational groups subdivided into two musical groups should be excluded. As stated by the more precise wording following this sentence, what is actually prohibited is rather overlaps.

¹⁶As the reader can see, discussing these questions without the relevant formalism of rooted tree soon gets tricky, and this is why I shall introduce it in the next section.

¹⁷As explained above, synchronised videos provide good examples of possible denotations for a musical snippet, that can be — unlike mental *imagery* proper — explicitly and extensively discussed.

¹⁸The excerpt can be watched here: <https://www.youtube.com/watch?v=H19jByxrqlw>.

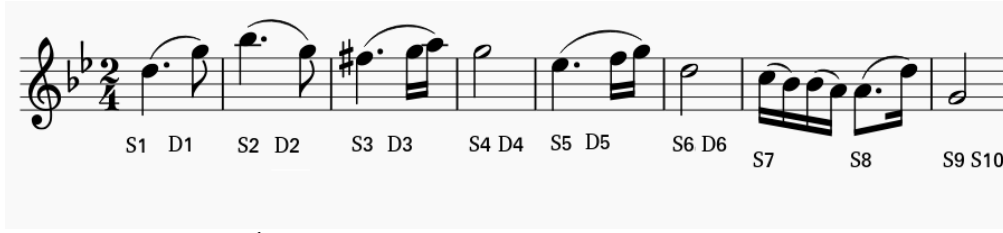


Figure 6: Theme from 5th Hungarian Dance, bars 17–24. The letters represent Chaplin’s actions

Ds are on the offbeats rather than on the beats, given that they are shorter, “pointed” actions, whereas the Ss are horizontal and sustained, and therefore more in line with the dotted quarter notes in the melody. This is in line with Schlenker’s theory on iconicity and stability.

Considering structures, on the situation side, giving a micro-analysis event by event is too unreliable in the absence of precise parsing rules like the ones we know for music (see below), but we can speculate at least three possible possible ways of forming larger groups:

1. According to Gestalt principles there may be three larger groups, one from S_1 to S_4 when the barber spreads from left to right, one from D_4 to S_6 when he spreads from right to left, and the last one from D_6 to S_{10} when spreading speed doubles¹⁹. At top-level, the middle-group is then grouped together with the first one, considering the change in speed as most important, as shown here:

$$\left[[S_1 - S_4][D_4 - S_6] \right] \left[D_6 - S_{10} \right]$$

This is what I will refer to as the *functional parsing* of the shaving scene in the rest of the discussion, because the last group has a different function (since the barber does not add more foam on these spreads but rather levels it on the customer’s face).

2. We can consider a second parsing that is quite similar, the only difference being that the middle-group is grouped with the third rather than the first, for reasons of symmetry:

$$\left[S_1 - S_4 \right] \left[[D_4 - S_6][D_6 - S_{10}] \right]$$

This is what I will refer to as the *symmetric parsing* of the shaving scene.

3. However, another grouping structure, correctly perceived by an anonymous reviewer watching the video *without music*, may hold. This second analysis is based on the position of Chaplin’s feet: he first moves from left to right (on S_3), then from right to left (on D_5). These movements may be considered as group borders:

$$\left[S_1 - D_2 \right] \left[S_3 - S_5 \right] \left[D_5 - S_{10} \right]$$

This is what I will refer to as the *positional parsing* of the shaving scene. Here again, the middle-group may be grouped with the first or the third but the subsequent discussion does not need to distinguish between these two options.

¹⁹ D_4 is part of the second group since it is a preparation for S_5 , as a finer analysis would probably give D_1 grouped with S_2 and so on.

Other parsings might also be defensible, and it is only my intuition that leads me to prefer the first one. Experimental results such as those obtained by Zacks et al. (2001) would allow for a more reliable analysis. But the point I want to make here does not rest exactly on the three parsings above and would lead to the same conclusion with different candidates.

On the musical side, several grouping structures are competing with each other according to Lerdahl & Jackendoff's preference rules. To begin with, second beats may be grouped either with the preceding or the following first beat (if they are heard as upbeats, for example). On a larger scale of bars, the following two grouping structures are possible²⁰:

1. The most prominent interpretation rests on the parallelism preference rule (here harmonic parallelism essentially) as well as on the symmetry preference rule. It gives couples of bars, grouped together, as shown here (bars are renumbered from 1 to 8 for the sake of simplicity):

$$\left[[1 - 2][3 - 4] \right] \left[[5 - 6][7 - 8] \right]$$

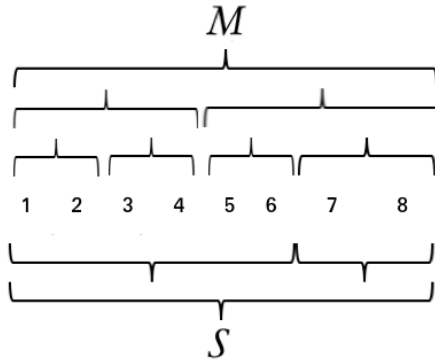
As I have just explained, the second beat of each even-numbered bar may be part of the next group, but I shall put this question aside here.

2. Another possible interpretation, though less convincing than the previous one, gives bars 1 to 6 grouped together against bars 7-8, with groups of two bars inside the first large group:

$$\left[[1 - 2][3 - 4][5 - 6] \right] [7 - 8]$$

What supports this analysis is mostly the change preference rule, resting on the fact that the rhythm and energy of bars 7-8 is different from bars 1-6.

Most preference rules conspire together to support the first grouping structure. We can see that it perfectly matches²¹ with the symmetric parsing of $\mathcal{S}$. If we then consider the functional analysis of $\mathcal{S}$ (first in the list above), it gives rise to the following diagram:



It is obvious in this diagram that the structures of $\mathcal{M}$ and $\mathcal{S}$ do not perfectly match. Actually, they scarcely ever match; at least according to Condition 4.1: since there is a group in $\mathcal{M}$ that contains events from bar 7 together with events from bars 5 and 6, which are grouped

²⁰As Lerdahl et Jackendoff (1983, §3.3) explain it is often the case that several grouping structures are competing each other.

²¹Except for the issue of offbeats mentioned above, which is secondary.

in $\mathcal{S}$ with events from bars 1-4, then this group of $\mathcal{M}$ should contain all these musical events from bars 1-4 too, which is not the case. Yet, this functional parsing seems the most consistent according to my intuition. If one wants to preserve some form of parallelism requirements, which seems necessary according to both our intuition when watching the video and above all to the previous examples I have given, several amendments are worth considering:

1. A first explanation is that the parallelism conditions may prevail only at low levels of grouping structures, that is, for short time spans. On the scale of several bars, their influence might fade. This would be accounted for by cognition effects: on the scale of a whole piece, it is well-known that even a trained listener cannot grasp the exact structure of, say, a movement of some Bruckner symphony. One may argue that listeners are only able to parse grouping structures for a span of just a few bars, and therefore the structural conditions might prevail only for the lowest levels of grouping structures in $\mathcal{M}$ and $\mathcal{S}$. This would explain why the mismatch is not bothering the attendant in the Chaplin clip, since it is not even perceived.
2. Another plausible explanation could be a retroactive reevaluation of the situation parsing: as the video is watched, one parsing is preferred over the others, but that does not mean that the dispreferred ones are completely eliminated from the mental space. Thus, the situation may be reparsed to match better with the music analysis. The phenomenon is well-known in dynamic sentence processing; for example, garden-path sentences illustrate well how the parsing system can process sentences several ways in parallel and select in real-time the most probable solution. This has also been noted for dance-music interactions: Charnavel (2022) shows with an excerpt from *The Rite of Spring* that some aspects of the choreography can — or “are precisely meant to” — help disambiguate the metrical structure of the music. Having the same mechanism working here would be plausible: while Charnavel shows how, when several structures for music or dance are competing, matching conditions will disambiguate the resulting dance-music object into a single structure, the Chaplin example can show that the same phenomenon would take place with grouping structure rather than metrical structure. This would also explain why the positional parsing is more easily perceived when watching the situation without music than with it. This structure is so far removed from that of music that it is not even detected.

Both positional and functional parsings suggest that a certain level of mismatch might be tolerated between $\mathcal{M}$ and $\mathcal{S}$ in order for $\mathcal{M}$ to denote $\mathcal{S}$, and thus that parallelism conditions should be considered as preference rules in competition with others rather than strict rules. Nevertheless, this tolerance for a certain degree of mismatch is undoubtedly higher in artistic creations (the main source of our examples, *e.g.* dance or videos) than in the mind of a listener who processes music and constructs more ‘literally’ their own meaning $\mathcal{S}$. We then get what could be compared, in literature, to a metaphor: the artistic interest lies in the fact that the signified does not correspond exactly, in one way or another, to the signifier. Therefore, is this certain degree of mismatch tolerated *de jure*, or is this transgression only tolerated as an artistic expression? It is hard to know at this stage of research. The fact remains that parallelism conditions play a role in the $\models$ relation and that this role is not necessarily as simple as mere identity between the two structures.

As explained above, there are several formal conditions that would exclude the unwanted mismatches. I will now turn to the formalisation of grouping structures, and then to an exploration of a range of those conditions.

5 Formalising grouping structures

In this section, I propose a formal framework in which the structural conditions developed above can be easily stated. As I have explained, these requirements relate to a certain matching between two structures: one for the musical part and the other for the situation part. So far we have imagined these structures as rooted trees, just like figure 4 shows, but the traditional formalism that had been developed (for music only) by Lerdahl et Jackendoff (1983) was in terms of “grouping structures”, a device that can easily be shown to be in a one-to-one correspondance²² with rooted trees, but whose drawback is that it has not been implemented in classical set theory as far as I know. In the aforementioned article on walk-denoting music (Migotti et Zaradzki, 2019), Migotti and I had discussed theoretical structural questions in the language of classical rooted trees²³. This had the advantage of allowing them to express a notion of event preponderance through the use of *labeled* trees, which are a classical variant of simple trees. However, the notion of tree embedding (from $\mathcal{S}$ to $\mathcal{M}$) that is formulated in this article does not exactly match, as I shall explain below, with the intuitions developed in the previous section, and especially the “no musical boundary inside situational groups” tenet. In this section, I shall thus propose an alternative, non-traditional notion of rooted trees, in which the statement and discussion of structural conditions on $\models$, as developed in section 6 below, is more precise and relevant.

This new framework, in which trees are no longer defined as particular kinds of graphs with specific properties (connexity, acyclicity and so on) as they usually are, is inspired from Stabler (1997), who developed it to study linguistic matters. Here trees are not conceived as vertices with edges, but as vertices equipped with a domination relation. More precisely:

Definition 1. A *rooted tree* (or a *tree*) is a couple $(V, \triangleleft^*)$, where:

1. V is a finite set, whose elements are called the tree’s *vertices*;
2. $\triangleleft^*$ is an order relation on V , called *domination relation*, such that :
 - (a) $\triangleleft^*$ has a greatest element (that is, an element x such that for every $y \in V$, $x \triangleleft^* y$), called the tree’s *root*;
 - (b) for all vertices x, y, z , if $y \triangleleft^* x$ and $z \triangleleft^* x$, then $y \triangleleft^* z$ or $z \triangleleft^* y$.

$y \triangleleft^* x$ is to be read “ y dominates x ”, or “ x is dominated by y ”. The notation is explained by the fact that a tree’s visual representation often has the shape of an isocles triangle, the dominating vertices being at its top. Other wordings include: “ y is an ancestor of x ”.

Notation 5.1. If x dominates y and $x \neq y$, we shall write $x \triangleleft^+ y$, and talk about strict domination. The set of leaves (that is, vertices without descendant) of a tree $T = (V, \triangleleft^*)$ is written down L_A .

Let $\mathcal{M} = (m_1, \dots, m_n)$ be a musical piece. $\mathcal{M}$ is to be represented as a rooted tree with leaves $m_1, \dots, m_n$, and whose subtrees will give the grouping structure. The trees we have defined above are not horizontally ordered: one may switch two sister branches without changing the tree. For instance, figure 7 shows two representations of the same tree.

²²Modulo some exceptions as overlaps and occlusions, discussed by Schlenker (2019, Appendix 3).

²³That is, a tree is a couple of sets (V, E) , V being the set of vertices and E that of edges.

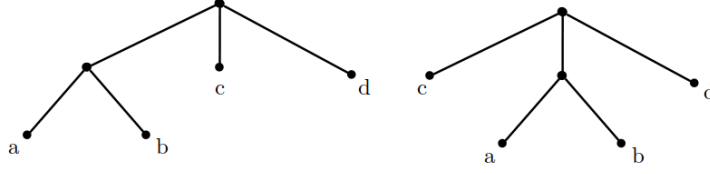


Figure 7: Two representations of the same tree

Figure 8: Example of grouping structure for Mozart's 40th Symphony, reproduced from Lerdahl et Jackendoff (1983, chap. 3)

Since our trees' leaves are intended to represent musical events and since musical events are ordered in time, we will need to distinguish these two representations. We therefore introduce a total ordering $\preceq$ on leaves, with the following visual convention: if $x \preceq y$, x is to be represented on y 's left. From now on every tree will be considered together with such a total order on its leaves.

In order to avoid crossings, we shall ask moreover that subtrees' leaves form intervals: if B is a subtree in A , $a \preceq b \preceq c$ are leaves in A , and $a, c \in L_B$, then $b \in L_B$ too.

Our first goal is to show that these ordered trees implement Lerdahl and Jackendoff's grouping structures. For this implementation to be one-to-one, we need to exclude vertices of degree 1 from our trees. We shall thus add to our definition that every vertex that is not a leaf has at least two children.

Lerdahl and Jackendoff are not explicit when it comes to the definition of their grouping structures on a musical piece $\mathcal{M} = (m_1, \dots, m_n)$ — they only *draw* them, as on figure 8 reproduced from their book — but the most natural way to define them is as a set $\mathcal{G}$ of subsets of $\{m_1, \dots, m_n\}$, each of these subsets representing a bracket, such as $\{m_1, m_2, m_3\}$ for the first one on figure 8. These subsets must verify a certain number of properties in order to be called grouping structures, and these properties are exactly what Lerdahl and Jackendoff call grouping well-formedness rules. They state five of them, the first one being, for instance, that “any contiguous sequence of pitch-events, drum beats, or the like can constitute a group, and only contiguous sequences can constitute a group”, which we could rephrase here just as “subsets in grouping structures must be intervals”. Moreover, since subsets of $\{m_1, \dots, m_n\}$ are obviously in a one-to-one relation with subsets of $\{1, \dots, n\}$, we can get rid of the m 's. Rephrasing Lerdahl and Jackendoff's five well-formedness rules in this formalism, we get:

Definition 2. A *grouping structure* on $\{1, \dots, n\}$ is a set $\mathcal{G} \subset \mathcal{P}(\{1, \dots, n\})$ (whose elements $G \in \mathcal{P}(\{1, \dots, n\})$ are called *groups*) such that:

1. Every $G \in \mathcal{G}$ is an interval (i.e.: $\forall i, j \in G, i < j \Rightarrow (\forall k \in \{1, \dots, n\} \text{ s.t. } i < k < j, k \in G)$),
2. $\{1, \dots, n\} \in \mathcal{G}$, $\emptyset \notin \mathcal{G}$ and $\forall i \in \{1, \dots, n\}, \{i\} \in \mathcal{G}$.

3. $\forall G_1, G_2 \in \mathcal{G}, G_1 \cap G_2 \neq \emptyset \Rightarrow (G_1 \subset G_2 \text{ or } G_2 \subset G_1),$
4. $\forall G_1, G_2 \in \mathcal{G}, G_2 \subsetneq G_1 \Rightarrow \exists k \in \mathbb{N}^*, \exists H_1, \dots, H_k \in \mathcal{G} \text{ s.t. } G_1 = G_2 \sqcup H_1 \sqcup \dots \sqcup H_k.$

In addition to the first rule discussed above, the second rule states that — in Lerdahl&Jackendoff’s words — “a piece constitutes a group”²⁴, the third one that “if a group G_1 contains part of group G_2 , it must contain all of G_2 and the last one that “if a group G_1 contains a smaller group G_2 , then G_1 must be exhaustively partitioned into smaller groups”. Where is the fifth rule, shall ask an attentive reader? It just states that “a group may contain smaller groups”, which is already contained in our using $\mathcal{P}(\{1, \dots, n\})$ ²⁵.

Based on this definition, it is trivial to show that if $\mathcal{G}$ is a grouping structure, then $(\mathcal{G}, \supseteq)$ is a rooted tree in the sense of Definition 1: $\{1, \dots, n\}$ is the greatest element and the property 2(b) is ensured by rule 3 from Definition 2. Moreover, the fourth well-formedness rule says that $\mathcal{G}$ has no vertex whose degree is equal to 1. The leaves in $\mathcal{G}$ are obviously the singletons.

Conversely, a given labeled rooted tree will give rise to a unique grouping structure, which means that this implementation of Lerdahl and Jackendoff’s grouping structures as rooted trees without vertex of degree 1 is actually one-to-one, in the following sense:

Proposition 1. Let $A = (V, \triangleleft^*)$ be an ordered rooted tree without vertex of degree 1. Suppose that the leaves of A are numbered from 1 to n and for every leaf $x \in L_A$, write $f(x) = \{i\}$, where i is the number associated with x . Then there exists a unique grouping structure $\mathcal{G}$ on $\{1, \dots, n\}$ such that f can be extended into a tree isomorphism from A to $(\mathcal{G}, \supseteq)$, and this isomorphism is then unique too.

Proof. It is easy to extend f recursively into a map from V to $\mathcal{P}(\{1, \dots, n\})$: for every vertex $v \in V$, note $v_1, \dots, v_k$ its children²⁶, and let $f(v) = f(v_1) \cup \dots \cup f(v_k)$. Then by construction $f(v)$ is the set of all leaves dominated by v , and the following holds:

$$\forall v_1, v_2 \in V, f(v_2) \subseteq f(v_1) \Leftrightarrow v_1 \triangleleft^* v_2$$

Since $\triangleleft^*$ is antisymmetric, f is injective, and $\mathcal{G}$ can be defined as just $f(V)$, so that f becomes bijective. Now checking the definition shows that $(\mathcal{G}, \supseteq)$ is indeed a grouping structure here, and f is now an isomorphism. Existence is therefore established.

Unicity still needs to be. Let $\mathcal{G}$ be a grouping structure and g be an isomorphism from A to $(\mathcal{G}, \supseteq)$ that extends f . Let me show by induction that:

$$\forall v \in V, g(v) = \bigcup_{x \in L_B} f(x)$$

(where B is the subtree in A with root v) which will prove both the unicity of the grouping structure and that of the isomorphism. This result is trivial when v is a leaf in A . Otherwise let $x \in L_B$ be a leaf under v . Since g is an isomorphism, $f(x) = g(x) \subseteq g(v)$. Hence the inclusion $\bigcup_{x \in L_B} f(x) \subseteq g(v)$ holds. Conversely, let $y \in g(v)$. Since v is not a leaf, v has descendants,

²⁴I here add the fact that $\emptyset \notin \mathcal{G}$ that was not considered by Lerdahl&Jackendoff, and the fact that all singletons need to be in $\mathcal{G}$, which is not present in Lerdahl&Jackendoff either but which I need to establish the subsequent correspondence with rooted trees.

²⁵It seems natural to actually interpret this rule as stating that a group can contain *only* smaller groups, as opposed to a copy of itself. Here set theory’s axiom of extensionality helps, and there is no rule to add since our groups are defined as sets.

²⁶One says that y is a child of x if $x \triangleleft^* y$ and there is no $z \in V$ s.t. $x \triangleleft^* z \triangleleft^* y$ besides x and y .

and $g(v) \in \mathcal{G}$ has subgroups (those descendants' images). According to the fifth last point of Definition 2, $g(v)$ is composed of several subgroups, of one of which, say $H = g(v')$, y belongs, with $v \triangleleft^+ v'$. By induction hypothesis $g(v') = \bigcup_{x \in L_{B'}} f(x)$ (where B' is the subtree in A with root v'), so there exists $x \in L_{B'}$ such that $y \in f(x)$, and since $L_{B'} \subset L_B$, $y \in \bigcup_{x \in L_B} f(x)$. $\square$

This correspondance allows us to implement grouping structures as ordered rooted trees without vertice of degree 1. Another question is to know how a given musical piece is assigned a grouping structure: how does the parsing take place and by which rules is the grouping structure derived? Several systems have been proposed by Lerdahl et Jackendoff (1983) through preference rules, and I will not go into them here. Recall that the goal of the article is to give some tools to study parallelism between structures, no matter how these structures were obtained in the first place. We are now able to examine structural parallelism conditions in the formalism of trees.

6 Structural rules for the denotation relation

Back to musical semantics, let us now consider two rooted trees $\mathcal{S}$ and $\mathcal{M}$ (both identified with their vertices) representing the grouping structures of a situation and a musical piece. Since they are chronologically ordered and carry the same number n of leaves²⁷, let us note $e_1 \preccurlyeq \dots \preccurlyeq e_n$ the leaves of $\mathcal{S}$ and $m_1 \preccurlyeq \dots \preccurlyeq m_n$ those of $\mathcal{M}$. Let $f : L_{\mathcal{S}} \rightarrow L_{\mathcal{M}}$ be the leaf-mapping function defined by $f(e_i) = m_i$ for every i , and g its inverse function.

Recall that the basic intuition, already stated by Migotti et Zaradzki (2019), was that “a group on the situation side must not contain a group boundary on the musical side²⁸” — while a group on the musical side may or may not contain a group boundary on the situation side; this asymmetry is discussed below. My main goal in the rest of this article is now to find a formal condition that captures this intuition.

Condition 4.1 from Migotti et Zaradzki (2019), according to which the leaf-mapping function f can be extended to an embedding from $V_{\mathcal{S}}$ to $V_{\mathcal{M}}$, was meant to implement that same basic intuition according to which a group on the situation side must not contain a group boundary on the musical side. More precisely, this means that if $e_{i_1}, \dots, e_{i_k}$ ($k \in \mathbb{N}^*, i_1 < \dots < i_k \in \{1, \dots, n\}$) and a are events of $\mathcal{S}$ associated with musical events $m_{i_1}, \dots, m_{i_k}$ and b , and if there is a group to which $e_{i_1}, \dots, e_{i_k}$ belong but a does not belong, then every musical group to which b with some of the m_{i_j} belong, must contain all the m_{i_j} (otherwise this would put up a boundary between the m_{i_j} where there was not between the e_{i_j}). In terms of trees: if $e_{i_1}, \dots, e_{i_k}$ belong to a subtree A of $\mathcal{S}$ while a does not belong to A , then every subtree B of $\mathcal{M}$ to which $b = f(a)$ and some of the $m_{i_j} = f(e_{i_j})$ belong, must contain all the m_{i_j} (that is: it must contain a copy of A). Reformulating again: if A is a subtree in $\mathcal{S}$ and B a subtree in $\mathcal{M}$, then: if $f(L_A) \cap L_B \neq \emptyset$ (that is, if some of the m_{i_j} belong to B) and if $L_B \not\subseteq f(L_A)$ (that is, if there exists $b \in L_B$ such that $a = g(b) \notin L_A$), then $f(L_A) \subseteq L_B$ (that is, $m_{i_1}, \dots, m_{i_k} \in L_B$), which is again equivalent to the most easily stated:

Condition 6.1. If A is a subtree in $\mathcal{S}$ and B is a subtree in $\mathcal{M}$, and if $f(L_A) \cap L_B \neq \emptyset$ then $f(L_A) \subseteq L_B$ or $L_B \subseteq f(L_A)$.

²⁷which is our working hypothesis here: the case where some events from $\mathcal{M}$ would refer to no event in $\mathcal{S}$ is also discussed by Migotti et Zaradzki (2019).

²⁸This wording is vague and less precise than the formal wording that follows. Indeed, it suggests that situational groups subdivided into two musical groups should be excluded. What is prohibited by the formal condition is rather overlaps, where a musical subgroup extends beyond one side of a situational group.

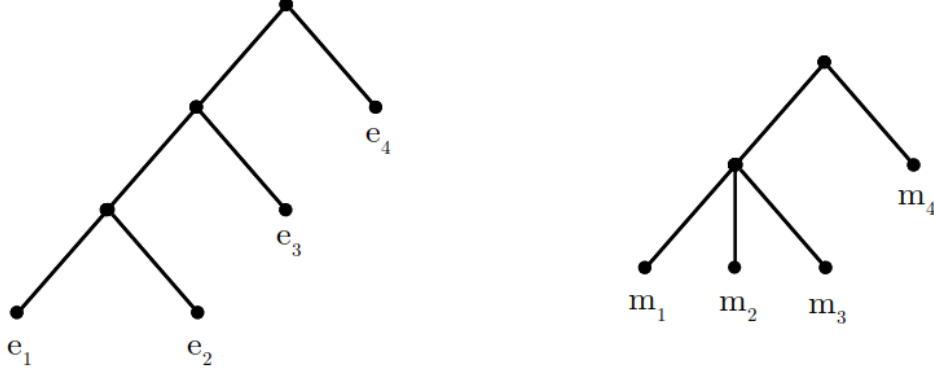


Figure 9: Condition 6.1 is satisfied by this pair of trees

The point of this condition is that $f(L_A)$ is not necessarily formed by the leaves of a *subtree* of $\mathcal{M}$, otherwise the condition would be automatically satisfied, since two subtrees of a tree are always one contained in another.

First let us note that this condition is not empty, in the sense that it would not force $\mathcal{S}$ and $\mathcal{M}$ to be identical, as can be seen on Figure 9, where one can exhaustively check that Condition 6.1 is satisfied by two trees that are different.

Second, condition 6.1 does rule out the problematic denotations as that on Figure 5 from *Fantasia 2000* (see section 4.1), which would survive if no structural condition held.

Now one may ask why this asymmetry in Condition 6.1, which states “no musical boundary inside a situation group” but does not require anything on situation boundaries in musical groups. One could well imagine the reverse condition:

Condition 6.2 (Reverse of Condition 6.1). If A is a subtree in $\mathcal{S}$ and B is a subtree in $\mathcal{M}$, and if $L_A \cap g(L_B) \neq \emptyset$ then $L_A \subseteq g(L_B)$ or $g(L_B) \subseteq L_A$.

In our article on walk-denoting music (Migotti et Zaradzki, 2019), we posited by introspection that situation boundaries inside musical groups were not as infelicitous as musical boundaries inside situation groups, and this is the origin of Condition 6.1. While the purpose of this article is not to decide whether or not structural parallelism conditions should or should not be asymmetric (this should be the subject of a whole experimental study), it is a fact that Conditions 6.1 and 6.2 are actually equivalent, as the following proposition shows:

Proposition 2. Conditions 6.1 and 6.2 are equivalent.

Proof. Since f and g are bijective and inverse to one another, the problem is symmetric and it is sufficient to prove only one implication. Let us thus assume Condition 6.1 and prove Condition 6.2. If $L_A \cap g(L_B) \neq \emptyset$, let a be an element in this intersection. Then $f(a) \in f(L_A)$ and $f(a) \in L_B$, and $f(L_A) \cap L_B \neq \emptyset$, and therefore Condition 6.1 ensures that $f(L_A) \subseteq L_B$ or $L_B \subseteq f(L_A)$.

If $f(L_A) \subseteq L_B$, let $x \in L_A$: $f(x) \in f(L_A) \Rightarrow f(x) \in L_B \Rightarrow x = g(f(x)) \in g(L_B)$; we thus have $L_A \subseteq g(L_B)$.

Now if $L_B \subseteq f(L_A)$, let $x \in g(L_B)$: $x = g(f(x))$ so by injectivity of g , $f(x) \in L_B \subseteq f(L_A)$, and by injectivity of f , $x \in L_A$; thus $g(L_B) \subseteq L_A$. $\square$

Stated differently: “no musical boundary inside situation groups” amounts to “no situational boundary inside musical groups”. This proposition proves that if we want our semantic model to be asymmetric on the constraints on $\mathcal{M}$ and $\mathcal{S}$, these conditions are not relevant.

In our article, Migotti and I also proposed a condition on $\models$ in terms of embeddings. What we call an embedding from one tree to another is a map on vertices that preserves the domination relations. More specifically:

Definition 3. Let $A = (V_A, \triangleleft_A^*)$ and $B = (V_B, \triangleleft_B^*)$ be two rooted trees equipped with total order relations on their leaves, both written $\preceq$. An *embedding* from A to B is a map $h : V_A \rightarrow V_B$ such that:

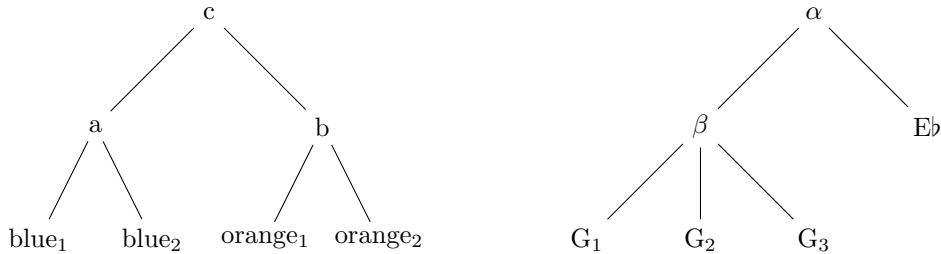
- (a) h is injective;
- (b) h is increasing on L_A to L_B
- (c) $\forall x, y \in V_A, x \triangleleft_A^* y \Rightarrow h(x) \triangleleft_B^* h(y)$

In the present article’s formalism, the embedding condition from [Migotti et Zaradzki \(2019\)](#) states that:

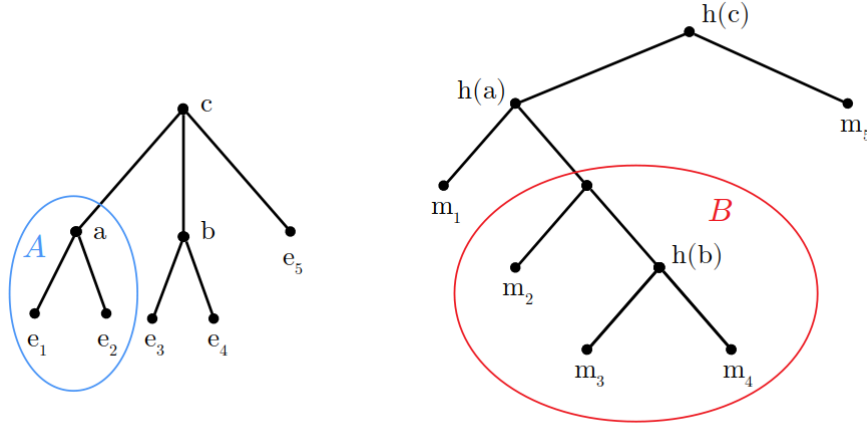
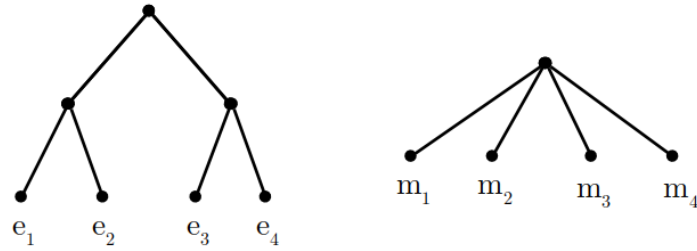
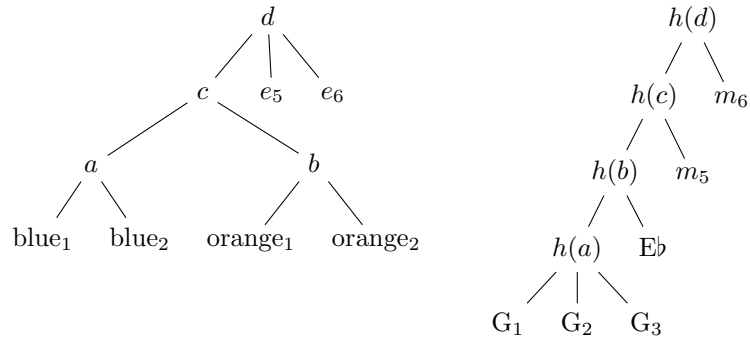
Condition 6.3. The leaf-mapping function $f : L_S \rightarrow L_M$ described above can be extended into an embedding $h : V_S \rightarrow V_M$.

While Migotti and I state that this is equivalent to Condition 4.1 (or equivalently, in our formalism, Condition 6.1), this is actually not the case. Figures 10 and 11 show that in fact, none of the two implications hold. On Figure 10, f can be extended to the embedding h as shown, but the drawn subtrees A and B violate Condition 6.1. Conversely, on Figure 11, $\mathcal{M}$ ’s structure is too poor both to violate Condition 6.1 and to authorise an embedding to exist (such an embedding would need at least two hierarchical levels, since it is injective).

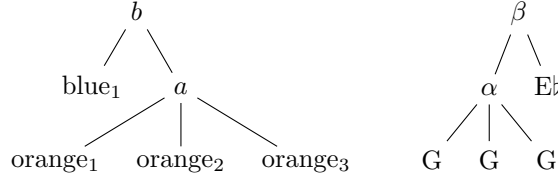
Therefore, both conditions do not express the same idea, but does Condition 6.3 still exclude the *Fantasia 2000* example? Well, it does, but not for relevant reasons:



Indeed, here no embedding from the left tree to the right tree can be found, for $\mathcal{S}$ has seven vertices while $\mathcal{M}$ has only six. But if we look at this snippet as a part of a larger picture, it is possible to build such an embedding, as shown here:

Figure 10: Counterexample to implication $\text{Cdt } 6.3 \Rightarrow \text{Cdt } 6.1$ Figure 11: Counterexample to implication $\text{Cdt } 6.1 \Rightarrow \text{Cdt } 6.3$ 

The map h described here is an embedding and Condition 6.3 is therefore not violated in this configuration, which nevertheless contains a copy of the mismatch from *Fantasia 2000*. While it is difficult to imagine a musical snippet that would give rise to this specific musical tree — and if corpus studies eventually prove that such a configuration never occurs, the definition of grouping structures needs to be refined — we should therefore consider Condition 6.3 with the highest suspicion, even if it does also exclude other unwanted mismatch configurations such as:



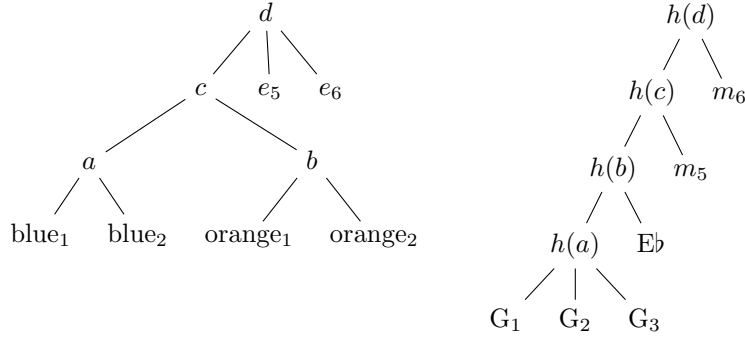
Here both $h(a)$ and $h(b)$ should dominate the Eb node, which is impossible if h is to be injective.

Let me now switch to the next condition, which is an improvement of Condition 6.3 able to deal with the previous examples:

Condition 6.4. The function $f : L_S \rightarrow L_M$ described above can be extended into an embedding $h : V_S \rightarrow V_M$ such that:

$$\forall x, y \in V_S, x \triangleleft^* y \Leftrightarrow h(x) \triangleleft^* h(y)$$

Obviously Condition 6.4 entails Condition 6.3. Now the following configuration repeated from above is excluded, as intended (as would be the one with $h(a)$ and $h(b)$ swapped around, for in S vertices a and b are not dominating each other):



Adding this reciprocal in Condition 6.4 makes it obviously stronger, such that Condition 6.4 now entails the first one:

Proposition 3. Condition 6.4 entails Condition 6.1, *i.e.* (mere repetition): if the function $f : L_S \rightarrow L_M$ can be extended into an embedding $h : V_S \rightarrow V_M$ such that:

$$\forall x, y \in V_S, x \triangleleft^* y \Leftrightarrow h(x) \triangleleft^* h(y)$$

then: for every subtree A of S and every subtree B of M , if $f(L_A) \cap L_B \neq \emptyset$ then $f(L_A) \subseteq L_B$ or $L_B \subseteq f(L_A)$

Proof. Assume that Condition 6.4 holds, and let A be a subtree of S and B a subtree of M such that $f(L_A) \cap L_B \neq \emptyset$. Note a the root of A and b the root of B . Consider an element $y_0 \in f(L_A) \cap L_B$, and $x_0 \in L_A$ such that $h(x_0) = y_0$.

Since $a \triangleleft^* x_0$, the domination $h(a) \triangleleft^* h(x_0) = y_0$ holds. Since we have $b \triangleleft^* y_0$ too in M , Definition 1 ensures that $h(a) \triangleleft^* b$ or $b \triangleleft^* h(a)$.

If $b \triangleleft^* h(a)$, then all the leaves of A have their images also dominated by $h(a)$ and then by b , so $f(L_A) \subseteq L_B$. Conversely, if $h(a) \triangleleft^* b$, let us show that $L_B \subseteq f(L_A)$. Let y be a leaf in L_B and let $x \in L_S$ be its antecedent: $x = g(y)$. Since $h(a) \triangleleft^* b \triangleleft^* h(x) = f(x)$, $a \triangleleft^* x$ according to the condition we have added, so x is a leaf in A , which is just what was needed. $\square$

Of course the reciprocal entailment Condition 6.1 $\Rightarrow$ Condition 6.4 cannot be true since Condition 6.1 did not entail the weaker Condition 6.3.

To finish, let me examine Condition 6.4 regarding the asymmetry issue. The existence of an embedding $h : V_S \rightarrow V_M$ such that: $\forall x, y \in V_S, x \triangleleft^* y \Leftrightarrow h(x) \triangleleft^* h(y)$ does not entail the existence of an embedding the other way around, for cardinality reasons again: h is injective but there may be more vertices in M than S total (if the grouping structure of M is more fine-grained, for example, while that of S is basic, *e.g.* one unique group) so that an injection from M to S would be impossible, whether it preserves domination or not. Thus Condition 6.4 is, strictly speaking, asymmetric. However, one may consider that the asymmetry of the system should not reside in cardinality reasons, which might appear to be insignificant. Yet when M and S , h would then be one-to-one and thus an isomorphism. This means that when V_M and V_S have exactly the same cardinal, the only way to fulfill Condition 6.4 is to make the structures of M and S identical.

This last constraint seems way too strict, but we should not forget that the cardinality of V_S is not that of L_S ; therefore, matching musical events with situation events (which was my working hypothesis in this article) does not entail that Condition 6.4 forces M and S to be isomorphic. In the end, I see no reason to *a priori* exclude Condition 6.4 on the mere ground of cardinality if one wants the system to be asymmetric, since cardinality is a good way to get asymmetry, and indeed is not unrelated to the matter of grouping structure: the more vertices the tree has, the richer the grouping structure it reflects. I conclude that if we are looking for asymmetry, Condition 6.4 could be a good candidate.

Moreover, Migotti and I evoked the hypothesis that not all musical events may have reference. This means that some leaves in M may not be mapped with leaves in S , but rather act as modifiers — or, if I may, adverbs — on the rest of the musical sentence. While first experiments²⁹ suggest it may not be so, if this is the case, the cardinality of V_M would in principle be bigger than that of V_S , which means: a) that isomorphism between M and S would be both impossible and undesirable; and b) that Condition 6.4 would then naturally give rise to asymmetry. Once future research has established whether or not perception obeys this asymmetry principle, it will provide a good criterion to decide which formal condition is the most susceptible to model the real syntax–semantics interface in music cognition.

7 Conclusions

In this article, I have argued, following an idea first suggested by Schlenker (2019) and then Migotti et Zaradzki (2019), that Schlenker’s rules of preservation of ordering are not sufficient to account for music semantics, and that structural rules are needed too. I have presented two elaborate musical examples supporting this: one from Migotti et Zaradzki (2019), the other one my own. I have then shown how syntactic information (here represented as grouping structures) on both music and situations can be formally implemented as rooted trees, and I have explored several conditions meant to capture the intuition that the music and situation structures should somehow match.

Two main formal conditions have been showcased, namely Condition 6.1 and Condition 6.4 (two more conditions have been examined: one is equivalent to Condition 6.1, while the other appears to be irrelevant). I have shown that they are not logically equivalent. Both conditions correctly rule out the mismatches I built up to show that rules of preservation of ordering were missing something. The main difference between them is that the former assumes a form of

²⁹Not published yet — personal communication

symmetry between the structures of music and situation, while the latter does not. To put it differently, while Condition 6.1 requires the music structure and the situation structure to share the same constraints, Condition 6.4 is more restrictive on the music structure. Lacking corpus or experimental data, both conditions remain plausible, and I will leave both possibilities open, which may eventually give rise to two different and consistent models, one symmetric and the other asymmetric.

8 Statements and declarations

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